Day 5 Deliverable: Baseline Methods, Comparison Criteria, Result Retrieval and Run Report

Insider Threat Detection System, Capstone Group 7 · 7 November 2026

1. Purpose

This document is the Day 5 deliverable. It answers both columns of the task table.

Column	Day 5 task	Where answered
Research and methodology	Define baseline methods or comparison criteria where applicable	Sections 2, 3 and 4
Prototype development	Implement result storage, retrieval, visualization or report generation	Section 5

The repository already stored results (results, runs, trained_models), served them through the API, and showed them in the dashboard (flagged list, session detail, user timeline, CSV download). Before Day 5 every comparison was between machine-learning variants. Day 5 adds the missing reference point, a comparison table with bars in the dashboard, and a printable run report. Evidence is in Section 6 and limits are in Section 7.

2. Baseline methods

The question a reviewer will ask is whether the models beat something simple. Four reference methods answer it. None of them learns anything. Each rule has one flag threshold, chosen on the validation part exactly like for the models, so the test part is never used for tuning.

Method	Definition	Why it is included
No skill	Flag every session. Precision equals the share of malicious sessions, recall is 1, and F1 is 2p/(1+p) for a share p.	The floor. Any useful method must be far above it.
Rule: after-hours session	Flag a session that started before 07:00 or after 19:00.	The simplest rule an analyst could write.

Method	Definition	Why it is included
Rule: after hours with a removable device or leak site	Flag a session that started after hours and also had a removable-device connection or a visit to a file-sharing or leak site.	A hand-written rule that combines the typical signs of data theft.
Rule: largest deviation from the user's own normal	The score is the largest of the 10 deviation features (how far the session is from the user's own running baseline). Flag at or above a threshold chosen on validation data.	A statistical rule that uses the personal baseline but no learning.

The earlier baselines stay: Logistic Regression as the simple supervised model, the feature groups, the combination rules and raw against calibrated scores (comparisons A to D in EVALUATION_CRITERIA.md). The new reference methods are comparison F. The letter E stays reserved for the planned transfer to SPEDIA, which has not been run.

3. Comparison criteria

Criterion	Measure	Reason
Main score	F1 on the held-out test part, with precision and recall	Less than 1% of sessions are malicious, so accuracy would hide the problem.
Ranking quality	Average precision	It does not depend on one threshold.
Analyst workload	False alarms per 1,000 benign sessions	It shows what a flag costs.
Probability quality	Brier score, models only	A rule gives no probability.
Same ground	The same time-ordered test part for every method. Thresholds are chosen on validation data only.	Nobody gets an advantage from the test labels.

Criterion	Measure	Reason
Stability	Mean and range over five generated data sets (seeds 7 to 11) for comparison F; one data set in the dashboard	One data set can be lucky.
Decision rule	A method clearly beats another when its lowest F1 over the five data sets is above the other's highest	Overlapping ranges are not called a win.

4. Results

4.1 Comparison F, five generated data sets (mean, lowest to highest)

Run on a Windows computer on 7 November 2026 with Python 3.14.5. numpy, scikit-learn, xgboost and shap have the same versions as on the first computer; pandas is 3.1.1 instead of 3.0.6.

Method	F1	Average precision	Precision	Recall	False alarms per 1,000 benign
Combined risk score (all 39 features)	0.91 (0.82 to 1.00)	0.99	0.96	0.88	0.7
Rule: after hours with a removable device or leak site	0.49 (0.38 to 0.61)	0.33	0.93	0.34	0.4
Rule: largest deviation from the user's own normal	0.43 (0.31 to 0.52)	0.42	0.47	0.44	8.5
Rule: after-hours session	0.27 (0.18 to 0.36)	0.09	0.23	0.34	16.4
No skill: flag every session	0.03 (0.02 to 0.03)	0.01	0.01	1.00	1,000

4.2 One retrained run in the dashboard (default data, seed 7)

Method	Precision	Recall	F1	Average precision	Brier	Threshold
Combined risk score	0.80	1.00	0.89	1.00	0.001	8.1
Logistic Regression	1.00	1.00	1.00	1.00	0.000	0.8
XGBoost	0.94	1.00	0.97	1.00	0.001	86.0

Method	Precision	Recall	F1	Average precision	Brier	Threshold
Isolation Forest	0.83	0.63	0.71	0.73	0.006	28.4
Rule: largest deviation from the user's own normal	0.41	0.44	0.42	0.36	-	5.1
Rule: after hours with a removable device or leak site	1.00	0.31	0.48	0.32	-	1.0
Rule: after-hours session	0.22	0.31	0.26	0.08	-	1.0
No skill: flag every session	0.01	1.00	0.03	0.01	-	-

4.3 What the results show, and what they do not

- **The models clearly beat every simple method.** The combined score's lowest F1 (0.82) is above the best rule's highest (0.61), so by the decision rule it never lost to a rule on these data sets.
- **The hand-written rule is clean but narrow.** It raises 0.4 false alarms per 1,000 benign sessions, close to the combined score's 0.7, but it finds only about a third of the malicious sessions, while the combined score finds 88%.
- **The no-skill line is the floor.** Its F1 of 0.03 is set by the share of malicious sessions.
- **The combination does not beat the best single model.** In the dashboard run, Logistic Regression reaches F1 1.00 and XGBoost 0.97 against 0.89 for the combined score. In the five-data-set mean, Logistic Regression and XGBoost both reach 0.94 against 0.91. The test part of the dashboard run has about 16 malicious sessions, so one or two sessions move an F1 value a lot. The combined score still gives one calibrated 0 to 100 number with a stored explanation, but it should not be described as the most accurate model.
- **Two computers gave different XGBoost results.** Logistic Regression (0.94) and Isolation Forest (0.75) gave the same means and ranges as on the first computer. Only the results that depend on XGBoost changed (XGBoost 0.94 instead of 0.98, combined score 0.91 instead of 0.95). XGBoost is not guaranteed to give identical results on different hardware or operating systems. The cause was not tested.
- **All data is generated.** These numbers say nothing about real logs.

5. Prototype: result storage, retrieval, visualization and report

5.1 Storage and retrieval

What	Where it is stored	How it is retrieved
One result per session with risk score, flag and explanation	results collection	GET /api/sessions, GET /api/sessions/{id}, GET /api/users/{id}; CSV download of the flagged list in the dashboard
One record per run, with test results per model and per reference method	runs collection and the stored models	GET /api/run/latest; the run table in the dashboard
A printable report of the newest run	Generated on request, not stored	GET /api/run/report

5.2 Reference methods in the engine

A new module, itd/rules.py, holds the three rules and the no-skill line. It is used by the engine and by the experiment script, so the numbers cannot drift apart. When models are trained, the engine evaluates the reference methods on the test part and stores the results with the other test metrics. A run that reuses saved models shows the results stored at training time. A run saved before Day 5 has no reference rows, and the dashboard says to run again with Train new models.

5.3 Visualization

The run table in the dashboard now shows the four models, then a divider, then the reference methods. Every row has an F1 bar. A sentence under the table states the combined score, the best rule and the no-skill value. The risk bars in the flagged list, the explanation of a session and the user timeline already existed.

5.4 Run report

The button Download run report in the dashboard fetches GET /api/run/report and saves a file run-report.html. The page has four parts: the run summary (sessions, flagged, threshold, weights, run time, validation result), the test results with the reference methods and a comparison sentence, the 10 highest-risk flagged sessions with up to three plain-words reasons each, and a limits statement. It opens in any browser; Ctrl+P and Save as PDF gives a three-page PDF.

Security decisions: every value from the database is escaped before it goes into the HTML; the page has no scripts and no external files; the response is a download with no caching; any signed-in user may read it (it contains what the dashboard already shows), and each download is logged as report_viewed in the access log.

5.5 A defect found by comparing the two outputs

Isolation Forest's recall showed 0.63 in the dashboard and 0.62 in the report. The value is exactly 0.625. JavaScript rounds a tie up, and Python rounds it to the nearest even digit. The report now rounds half up with exact decimal arithmetic, like the dashboard. The fix is covered by test_numbers_round_like_the_dashboard, which also compares Python's results with the known JavaScript results for 0.625, 0.375, 1.005 and 19.25.

5.6 Tests and code changes

Tests: 74 passed and 1 skipped (the test that needs a real MongoDB server), up from 68 before the report and 63 before the reference methods. New tests: 5 in test_baselines.py (rule definitions, thresholds from validation data only, the no-skill values, storage by the engine) and 6 in test_report.py (content, escaping of hostile text, old runs and empty lists, rounding, and the download endpoint with the top 10 of the newest run only). One existing test was corrected because it assumed that every stored test result belongs to a model.

File	Change
backend/itd/rules.py	New. The reference methods.
backend/itd/report.py	New. The HTML report builder.
backend/itd/engine.py	Stores the reference results with the training metrics
backend/itd/api.py, schema.py	Report endpoint; the log action report_viewed
backend/scripts/run_experiments.py	Comparison F
backend/tests/test_baselines.py, test_report.py, test_pipeline.py	New tests; one corrected test
frontend/src/components/RunPanel.jsx, api.js	Reference rows, F1 bars, comparison sentence, report button
docs	EVALUATION_CRITERIA, API, ARCHITECTURE, SECURITY, DATABASE, DATA_RULES, TOOLS_AND_ALGORITHMS

6. Evidence

Figure 1 shows the test run on 7 November 2026: 74 passed, 1 skipped.

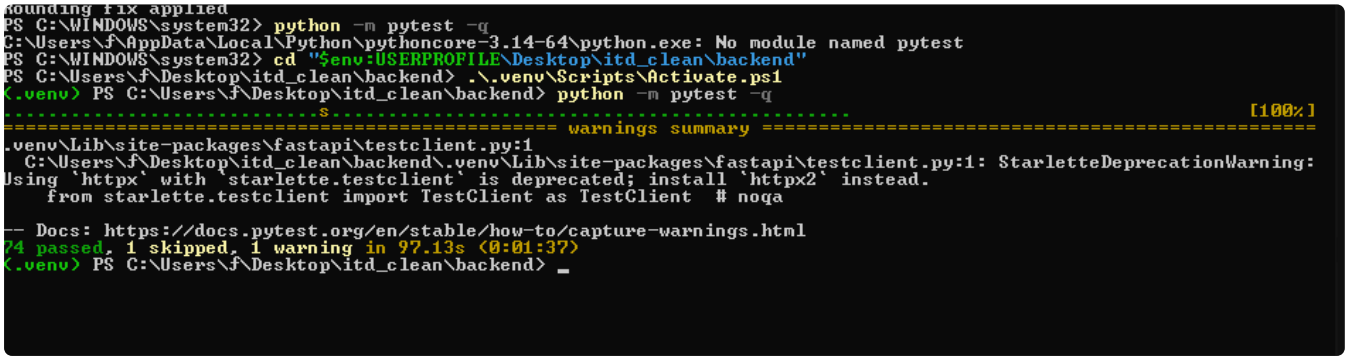


Figure 1. Test result.

Figure 2 shows the Run analysis card after a run with Train new models ticked. The status reads Finished with new models, in 12.7 seconds.

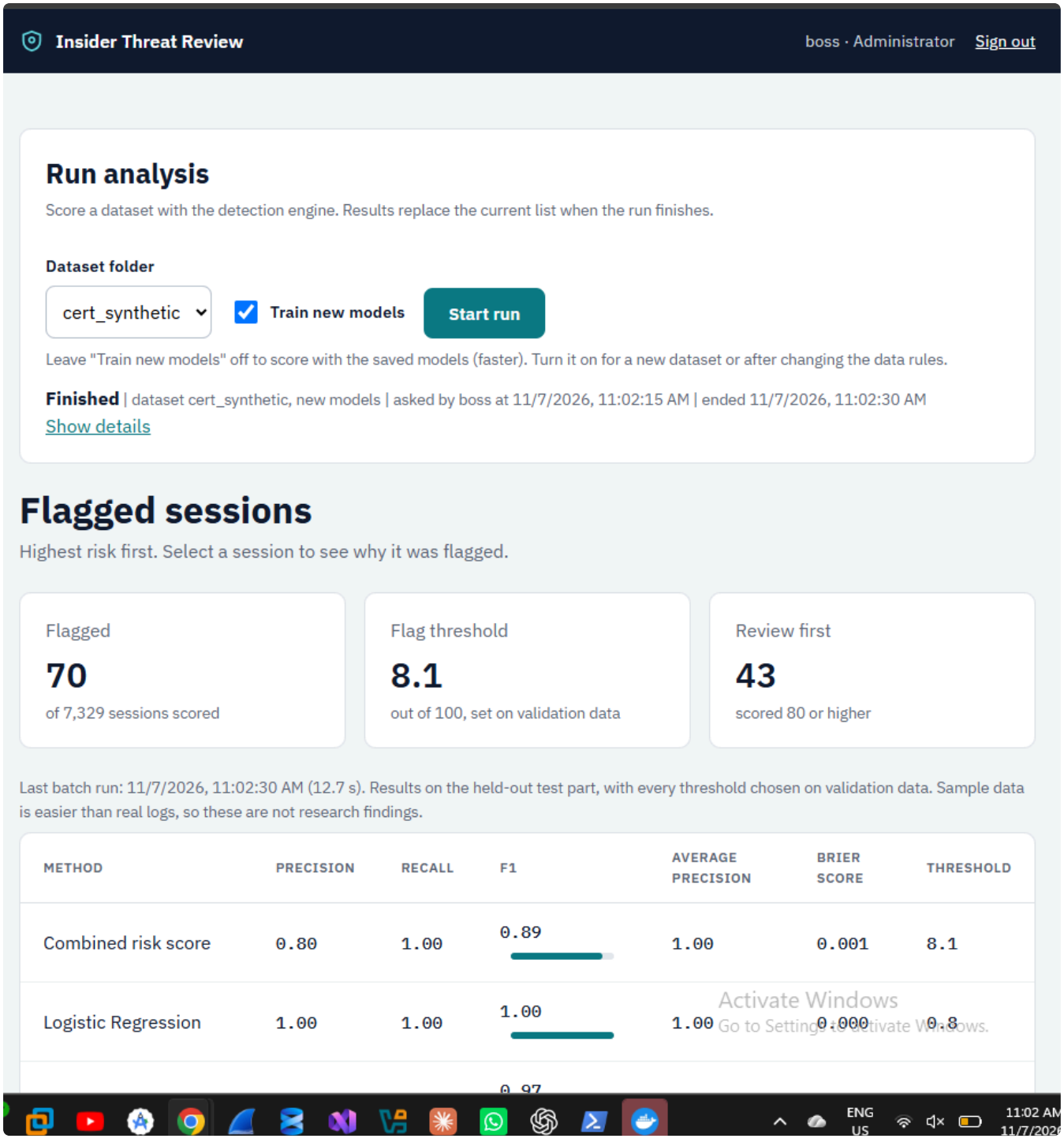


Figure 2. A finished training run started from the dashboard.

Figure 3 shows the run table after that run: the four models, the divider, the three rules and the no-skill row, each with an F1 bar, the comparison sentence and the Download run report button.

Last batch run: 11/7/2026, 11:02:30 AM (12.7 s). Results on the held-out test part, with every threshold chosen on validation data. Sample dat is easier than real logs, so these are not research findings.

METHOD	PRECISION	RECALL	F1	AVERAGE PRECISION	BRIER SCORE	THRESHOLD
Combined risk score	0.80	1.00	0.89	1.00	0.001	8.1
Logistic Regression	1.00	1.00	1.00	1.00	0.000	0.8
XGBoost	0.94	1.00	0.97	1.00	0.001	86.0
Isolation Forest	0.83	0.63	0.71	0.73	0.006	28.4
Simple reference methods (no learning; the threshold is chosen on validation data)						
Rule: largest deviation from the user's own normal	0.41	0.44	0.42	0.36	-	5.1
Rule: after hours with a removable device or leak site	1.00	0.31	0.48	0.32	-	1.0
Rule: after-hours session	0.22	0.31	0.26	0.08	-	1.0
No skill: flag every session	0.01	1.00	0.03	0.01	-	-

Brier score: lower is better (0 is perfect). Threshold is on the 0 to 100 scale for the models and in the units of each rule for the rules.

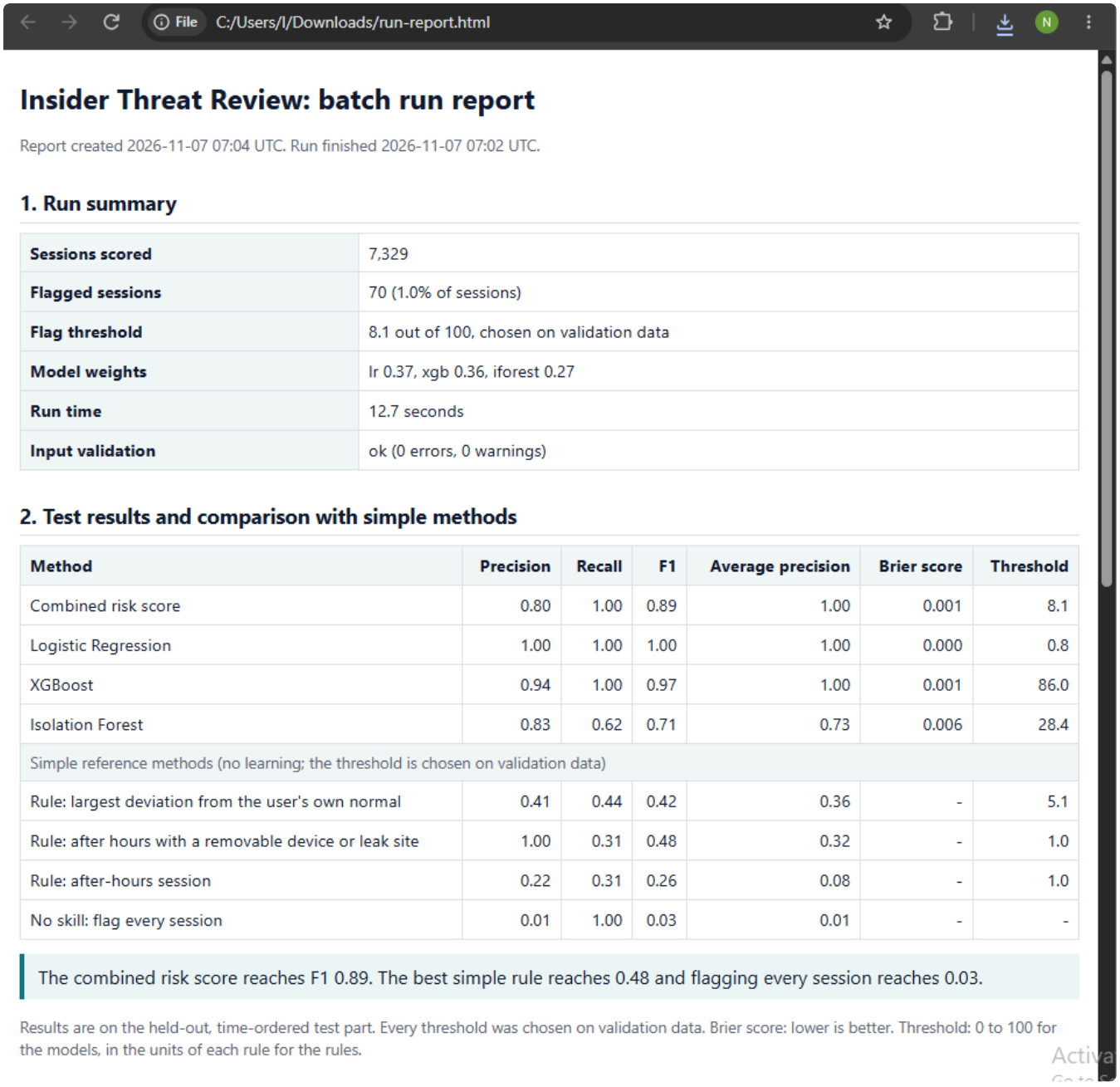
The combined risk score reaches F1 0.89. The best simple rule reaches 0.48 and flagging every session reaches 0.03.

Download run report One page you can open in a browser and save as PDF (Ctrl+P).

Activate Windows
Go to Settings to activate Windows.

Figure 3. Run table with the simple reference methods.

Figure 4 shows the top of the downloaded report: the run summary and the comparison table. It was taken before the rounding fix of Section 5.5, so Isolation Forest's recall reads 0.62 here.



Insider Threat Review: batch run report

Report created 2026-11-07 07:04 UTC. Run finished 2026-11-07 07:02 UTC.

1. Run summary

Sessions scored	7,329
Flagged sessions	70 (1.0% of sessions)
Flag threshold	8.1 out of 100, chosen on validation data
Model weights	lr 0.37, xgb 0.36, iforest 0.27
Run time	12.7 seconds
Input validation	ok (0 errors, 0 warnings)

2. Test results and comparison with simple methods

Method	Precision	Recall	F1	Average precision	Brier score	Threshold
Combined risk score	0.80	1.00	0.89	1.00	0.001	8.1
Logistic Regression	1.00	1.00	1.00	1.00	0.000	0.8
XGBoost	0.94	1.00	0.97	1.00	0.001	86.0
Isolation Forest	0.83	0.62	0.71	0.73	0.006	28.4
Simple reference methods (no learning; the threshold is chosen on validation data)						
Rule: largest deviation from the user's own normal	0.41	0.44	0.42	0.36	-	5.1
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No skill: flag every session	0.01	1.00	0.03	0.01	-	-

> The combined risk score reaches F1 0.89. The best simple rule reaches 0.48 and flagging every session reaches 0.03.

Results are on the held-out, time-ordered test part. Every threshold was chosen on validation data. Brier score: lower is better. Threshold: 0 to 100 for the models, in the units of each rule for the rules.

Figure 4. The run report, summary and comparison table.

Figure 5 shows the 10 highest-risk flagged sessions in the report, each with its reasons in plain words.

3. The 10 highest-risk flagged sessions

#	Risk	User	Session start	Minutes	Why it was flagged
1	95	U0073	2010-02-01 22:07 UTC	184	• Visited 6 file-sharing / leak sites (usual 0.0). • Activity rate was 9.1 events per hour. • Logged on at 22.1h (usual 10.0h).
2	95	U0027	2010-02-09 21:41 UTC	171	• Handled 24 archive (.zip) files (usual 0.0). • Activity rate was 15.7 events per hour. • Logged on at 21.7h (usual 9.3h).
3	95	U0087	2010-01-14 06:25 UTC	233	• Visited 5 file-sharing / leak sites (usual 0.0). • Activity rate was 6.7 events per hour. • Sent 9467 KB of email (usual 161 KB).
4	95	U0101	2010-01-28 21:51 UTC	216	• Visited 2 file-sharing / leak sites (usual 0.0). • Activity rate was 5.5 events per hour. • Logged on at 21.9h (usual 8.2h).
5	95	U0101	2010-03-26 06:26 UTC	98	• Visited 6 file-sharing / leak sites (usual 0.2). • Activity rate was 16.5 events per hour.
6	95	U0110	2010-01-25 05:45 UTC	299	• Visited 3 file-sharing / leak sites (usual 0.2). • Activity rate was 3.4 events per hour. • Sent 5 emails outside the company (usual 0.2).
7	94	U0080	2010-01-19 06:03 UTC	203	• Handled 23 archive (.zip) files (usual 0.0). • Activity rate was 13.6 events per hour. • Touched 25 files (usual 2.1).
8	94	U0110	2010-01-22 20:27 UTC	262	• Visited 2 file-sharing / leak sites (usual 0.0). • Activity rate was 4.3 events per hour. • Logged on at 20.4h (usual 10.2h).
9	94	U0011	2010-03-26 22:59 UTC	125	• Visited 5 file-sharing / leak sites (usual 0.3). • Activity rate was 12.9 events per hour. • Logged on at 23.0h (usual 9.2h).
10	94	U0011	2010-01-18 21:54 UTC	140	• Visited 6 file-sharing / leak sites (usual 0.8). • Activity rate was 9.8 events per hour. • Logged on at 21.9h (usual 8.9h).

Acti
Go to

The explanation lists what pushed the XGBoost score up and how far the session was from the user's own normal. It does not cover the other two models.

Figure 5. The 10 highest-risk sessions with their reasons.

Figure 6 shows the analyst view: the same results and no Run analysis card.

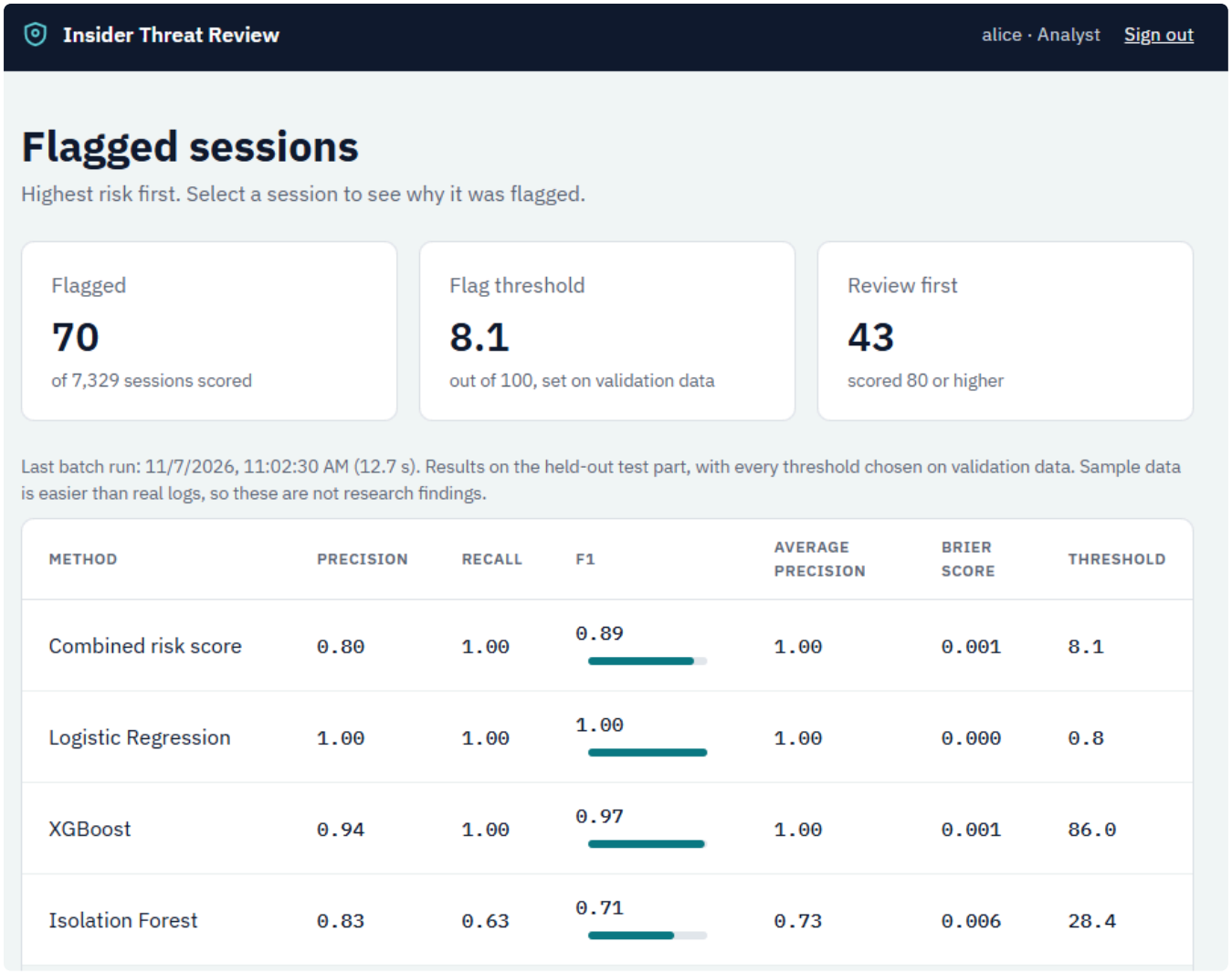


Figure 6. Analyst view.

Figure 7 shows the database check after the run: the sessions by part (train 5,130, validation 1,099, test 1,100), all eight methods stored, the report_viewed log entry and the finished request.

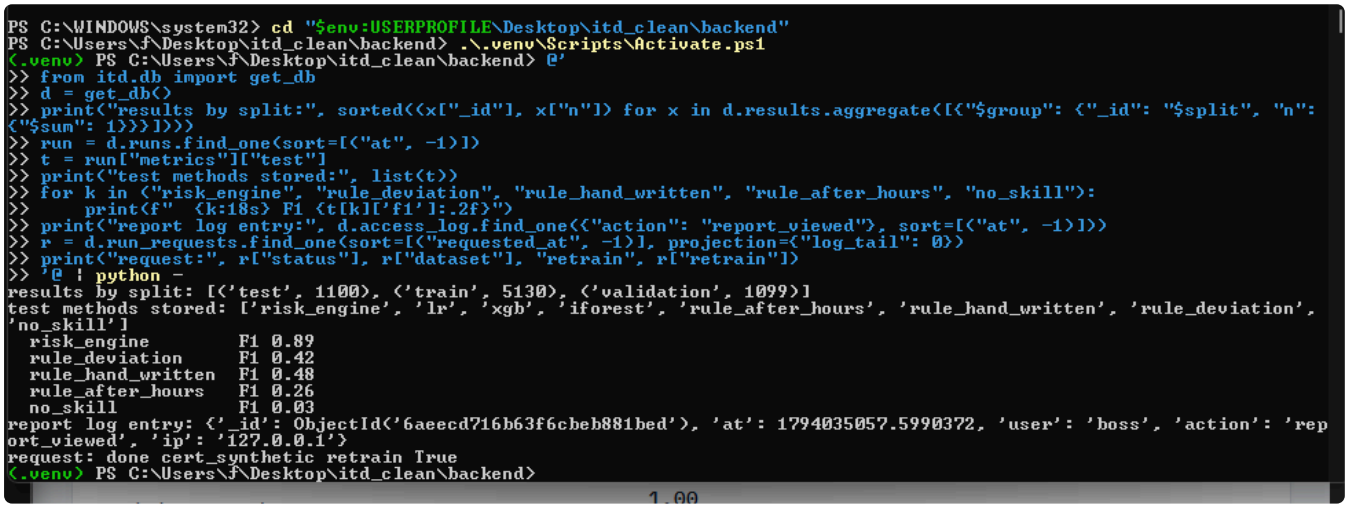


Figure 7. Records stored after the run.

Output of comparison F (python scripts\run_experiments.py --seeds 5, last block):

```
F simple reference methods
  combined risk score (all 39 features)  F1 0.91 (0.82 to 1.00)  AP 0.99
precision 0.96  recall 0.88  FP/1000 benign 0.7
  Rule: after-hours session              F1 0.27 (0.18 to 0.36)  AP 0.09
precision 0.23  recall 0.34  FP/1000 benign 16.4
  Rule: after hours with a removable device or leak site  F1 0.49 (0.38 to
0.61)  AP 0.33  precision 0.93  recall 0.34  FP/1000 benign 0.4
  Rule: largest deviation from the user's own normal  F1 0.43 (0.31 to 0.52)
AP 0.42  precision 0.47  recall 0.44  FP/1000 benign 8.5
  No skill: flag every session           F1 0.03 (0.02 to 0.03)  AP 0.01
precision 0.01  recall 1.00  FP/1000 benign 1000.0
```

7. Limits and open items

- The data is generated. The reference rules were written by the authors, who know the kinds of behavior in the data, so they are plausible analyst rules but not neutral ones. Other rules could score differently.
- The test part has few malicious sessions, so single F1 values move a lot. The five-data-set ranges show how much.
- The comparison between the two computers shows that XGBoost results can differ; the cause is not known.
- The report is HTML. A PDF comes from the browser's print function; the server does not create PDFs.
- The dashboard shows tables and bars, not precision-recall curves or score histograms. They were not built.
- The report was checked with generated data and in a browser. It was not checked with a screen reader or on a phone.
- Original CERT r4.2 and SPEDIA (comparison E) have not been run.